

VR Wayfinding Training for People with Visual Impairment using VR Treadmill and VR Tracker

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ABSTRACT

There are virtual reality (VR) wayfinding training systems for people with visual impairment, but there is a lack of studies about how training environments can affect spatial information acquisition of people with visual impairment. Using a VR treadmill and a VR tracker, we studied how walk-in-place and actual walking can affect the acquisition of spatial information with regard to paths and obstacles. Our results show that people with visual impairment remember routes better when trained with VR treadmill, but they remember obstacles better when trained with VR tracker. We evaluate the respective efficacies of these approaches on spatial information memorization.

Keywords: Virtual reality (VR), people with visual impairment, wayfinding, training.

Index Terms: Human-centered computing ~ Human computer interaction (HCI) ~ Empirical studies in HCI; Human-centered computing~ Human computer interaction (HCI)~Interactive systems and tools

1 INTRODUCTION

As the availability of virtual reality (VR) hardware devices such as VR treadmill and Vive tracker increases, technologies that allow people with visual impairment to access the safe and immersive wayfinding training environment are being developed [1-4]. As such, the VR helps people with visual impairment to safely build a cognitive map of new spaces without help of others [5]. A cognitive map is an image of the spatial structure of the environment and the spatial relationship of objects. When people with visual impairment form a cognitive map, they should be able to recall the relative distances and direction between landmarks, the route, obstacles, and objects they have experienced in a new space [6]. In this study, we examined the difference between VR wayfinding training while walking-in-place using a VR treadmill and VR wayfinding training while actually walking using a VR tracker.

The VR treadmill system can simulate an infinite range of virtual space even in a narrow physical space and it enables people with visual impairment to walk safely with support at the waist. The VR tracker system can track actual steps of users in large open spaces by putting markers on the body. These two systems are commonly used in VR walking systems for people with visual

impairment [1-4]. However, there is a lack of studies on how respective walking training methods affect the participant's capacity to memorize the spatial information of a walked environment. In our study, we have determined how well people with visual impairment remember the paths and obstacles after wayfinding training with these two walking simulation systems vis-à-vis treadmill and trackers.

2 IMPLEMENTATION

To achieve the goal of this study, a virtual wayfinding training system was developed for people with visual impairment. In this system, HTC Vive Pro and Unity3D 2019.2.14 versions were used. Walking VR environments were prepared with 3D spatial audio using the Google resonance audio SDK. Especially, two different walking environments were simulated for people with visual impairment. The first is a VR tracker environment in which people with visual impairment can actually walk on the floor with wearing Vive tracker. The second is a VR treadmill environment in which people with visual impairment can walk-in-place using Virtuix Omni. A Vive pro's head mounted device (HMD) was used in both environments to reflect the head movements of participants. HMD tracked the user's height, head position, and head direction to track the ear positions of participants, which is the arrival point of spatial sound. This supports the appropriate change of the direction and displacement of the environmental sound in VR. In addition, a virtual white cane to detect virtual obstacles was implemented by the VR controller. The controller also generates a vibration when the virtual cane hits a virtual object such as a wall or the obstacles. The implemented systems are shown in Figure 1.

3 EXPERIMENTS

We conducted two wayfinding training experiments with people with visual impairment. In the first experiment, we evaluated the memorization of the routes between with the VR treadmill and with the VR tracker. In the second experiment, we evaluated the memorization of the position of obstacles between with the VR treadmill and with the VR tracker.

In the first experiment, eleven participants with visual impairment were signed a consent form and answered demographic questions such as age and gender. The experiment was conducted using two VR wayfinding devices, the VR tracker and the VR treadmill. In the experimental procedure, all participants were demanded to wear both eye patches during the experiment to control errors caused by different eyesight. Before starting the experiment, the participants had 20 minutes of training to help them get used to each locomotion device. During the main experiment, each participant explored the five routes. Wayfinding training was conducted by walking the given routes with the guide dog. After the wayfinding training, 3D printouts of four different routes were presented to the participants from which participants chose the route they had walked. To minimize the learning effect, the order of the walking routes was changed by the Latin Square method

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Figure 1: The Virtu Omni (left) and HTC Vive Tracker (right), which are used in the experiments.

during the main experiment. The order of VR Treadmill or VR tracker in the experiment condition was also balanced by the Latin Square method. After completing the wayfinding training, the researchers interviewed the participants on what they felt their experience. The memorizing routes shows a significant difference between the VR tracker and the VR treadmill ($t(107.04)=1.980$, $p=0.05$) as Figure 2.

In the second experiment, sixteen participants with visual impairment were signed the consent form and wore eye patches on both eyes to control errors that could be caused by different eyesight. Participants had a 20-minutes training before the main experiment to get familiar with the training systems. During the experiment, each participant walked one path on the VR treadmill and another path with the VR tracker. The paths for each device were changed with the Latin Square. The order of experiencing devices was also balanced by the Latin Square method. The experimental task was to memorize the positions of the five obstacles. After the experiment, participants had to place the Lego blocks on the cloned route to indicate the locations of the identified obstacles. We collected data regarding how accurately participants positioned the Lego blocks in the routes. A Mann-Whitney U test, showed that there was a significant difference in the number of correctly recognized obstacles between the VR treadmill and the VR tracker ($U=2480$, $p<0.01$) as Figure 2. We found that the VR tracker environment allows people with visual impairments to remember obstacles better than the VR treadmill environment.

4 DISCUSSION

These findings can be related to cognitive maps for wayfinding by people with visual impairment. Route-level cognitive maps is one of the cognitive map which presents the route from departure to destination [6]. This cognitive map only have a summary understanding of the route which doesn't contain the detail map information like position of obstacles. We expect that people with visual impairment can make route-level cognitive maps by recognizing and memorizing directions while wayfinding training. VR treadmill allows user to feel the rotation of the entire torso because it allows user to walk in a circular structure. The result of our experiment showed that people with visual impairment memorized the route more correctly with VR treadmill than with VR tracker. On the other hand, a survey-level cognitive map is a representation of the spatial understanding of the overall environment. A person with only a route-level cognitive map choose from a variety of routes, but a person with a survey-level cognitive map can choose different routes [7]. One of the activities to develop a survey-level cognitive map is to investigate the path environment, such as looking for obstacles. The tracker might have an advantage in environmental investigations over treadmills because it allows the user to move the body freely making it easier to explore the surrounding environment. The result of our experiment also showed that people with visual

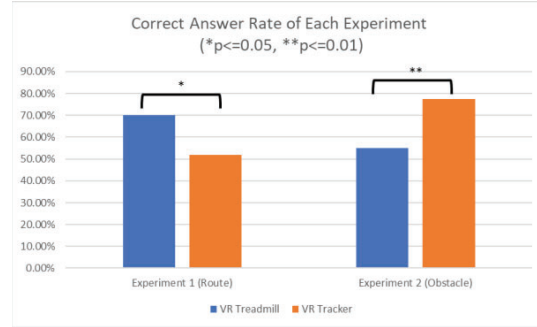


Figure 2: Correct answer rate from Experiment 1 and 2.

impairment memorized the position of obstacles more correctly with VR tracker than with VR Treadmill. So, the use of the VR tracker might be more suitable than the VR treadmill in investigating the obstacles on wayfinding training. We hope that our finding will benefit the assistive VR community.

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